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Final Project

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BUYING THROUGH AI AGENTS

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FINAL PROJECT DEFENCE

Meeting of the evaluating panel on this day, the student:

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Presented their final thesis on the following subject:

BUYING THROUGH AI AGENTS

At the end of the presentation and upon answering the questions of the members of the panel, this thesis was awarded the following grade:

--

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AI Usage Statement

The proposal for this research has been created with the support of Artificial Intelligence applications to assist with clarity, grammar, and formatting to enhance the proposal's purpose; however, the overall message, research design, research methodology, and research sources were created by the team responsible for developing this proposal.

Abstract

The way we shop is evolving due to artificial intelligence. Tools that previously only offered product recommendations can now compare possibilities, evaluate prices, and even execute transactions on our behalf, yet most customers are still reluctant to give up that degree of control. Understanding why is the central question this study seeks to answer. With an emphasis on the Spanish market, it specifically looks into how customers' readiness to embrace AI-based shopping agents is influenced by perceived usefulness, trust, perceived risk, and automation preference. Using survey data from 200 participants, five hypotheses are investigated based on an expanded version of the Technology Acceptance Model (TAM). Multiple Linear Regression, Random Forest validation, and K-means clustering are used in the study to determine the primary adoption determinants and segment consumers into meaningful profiles. The main adoption factors, according to the results, are perceived usefulness and trust; the multivariate model did not support perceived risk or the trust-risk moderation hypothesis.

Resumen

La forma en que compramos está evolucionando gracias a la inteligencia artificial. Aunque la mayoría de los clientes aún se resisten a renunciar a ese grado de control, las herramientas que antes solo ofrecían recomendaciones de productos ahora pueden comparar opciones, evaluar precios e incluso ejecutar transacciones en nuestro nombre. Entender por qué es la pregunta central que este estudio busca responder. Con énfasis en el mercado español, analiza específicamente cómo la disposición de los clientes a adoptar agentes de compra basados en IA se ve influenciada por la utilidad percibida, la confianza, el riesgo percibido y la preferencia por la automatización. Utilizando datos de encuestas de 200 participantes, se investigan cinco hipótesis basadas en una versión ampliada del Modelo de Aceptación de la Tecnología (TAM). En el estudio se utilizan la regresión lineal múltiple, la validación de bosques aleatorios y el agrupamiento K-means para determinar los principales determinantes de la adopción y segmentar a los consumidores en perfiles significativos. Los principales factores de adopción, según los resultados, son la utilidad percibida y la confianza; el modelo multivariante no respaldó el riesgo percibido ni la hipótesis de moderación confianza-riesgo.

Resum

La manera com comprem està evolucionant a causa de la intel·ligència artificial. Tot i que la majoria dels clients encara són reticents a renunciar a aquest grau de control, les eines que abans només oferien recomanacions de productes ara poden comparar possibilitats, avaluar preus i fins i tot executar transaccions en nom nostre. Entendre per què és la pregunta central que aquest estudi busca respondre. Amb èmfasi en el mercat espanyol, examina específicament com la disposició dels clients a adoptar agents de compres basats en IA es veu influenciada per la utilitat percebuda, la confiança, el risc percebut i la preferència d'automatització. Utilitzant dades d'enquestes a 200 participants, s'investiguen cinc hipòtesis basades en una versió ampliada del Model d'Acceptació de Tecnologia (TAM). A l'estudi s'utilitzen la regressió lineal múltiple, la validació Random Forest i l'agrupació de K-means per determinar els determinants principals de l'adopció i segmentar els consumidors en perfils significatius. Els principals factors d'adopció, segons els resultats, són la utilitat percebuda i la confiança; el model multivariant no va donar suport al risc percebut ni a la hipòtesi de moderació de la confiança-risc.

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Author Contributions

This thesis was developed jointly and equally by Clara Martin Peyrona and Alison-Rose Platt Portero. All sections of the document, including the literature review, methodology, data analysis, and discussion, were carried out collaboratively by both authors throughout the entire research process.

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Beyond just being an assignment, this thesis has provided us with experience working with real-world data; thinking critically and creatively; using a systematic approach to solving complex issues; and furthering our knowledge regarding both the scholarly method of research and the rapidly-evolving field of AI-driven ecommerce.

We hope our findings not only reflect what we have learned throughout this process, but also contribute meaningfully to the ongoing conversation around AI adoption in digital commerce.

Executive Summary

The rise of artificial intelligence in digital commerce has created a new category of technology: AI-based shopping agents capable of autonomously searching, comparing, and completing purchases on behalf of consumers, in a market expected to generate \$3 to \$5 trillion in global transactions by 2030 (Schumacher et al., 2026). While AI tools for product discovery and comparison are already widely used, with 73% of global consumers incorporating AI into their purchasing process, full delegation of purchasing authority to autonomous AI agents is still rare, with only 13% of consumers having completed a purchase directed by an AI agent. This disconnect between usage and adoption represents the central research problem of this study.

This thesis investigates the behavioural, psychological, and technological factors that influence consumers' willingness to adopt AI-based shopping agents, with a particular focus on the Spanish consumer context. Drawing on an extended Technology Acceptance Model (TAM), the study incorporates four independent constructs, perceived usefulness, trust, perceived risk, and automation preference, alongside adoption intention as the dependent variable, and tests five theoretically grounded hypotheses. An additional moderation hypothesis examines whether trust attenuates the negative effect of perceived risk on adoption intention.

The study employs a quantitative cross-sectional design, collecting data from 200 respondents via an online survey. A multi-method analytical framework is applied, combining Multiple Linear Regression (MLR) for hypothesis testing, Random Forest regression for non-linear validation, and K-means Clustering for consumer segmentation. This Business Intelligence approach enables both explanatory and predictive insights into adoption behaviour.

The empirical results reveal that Perceived Usefulness ($\beta = +0.392$, $p < .001$) and Trust ($\beta = +0.364$, $p < .001$) are the two primary drivers of adoption intention, together explaining 57.1% of the variance ($R^2 = .571$). Perceived Risk moved in the expected negative direction but did not reach statistical significance in the multivariate model, suggesting that usefulness and trust perceptions may override risk-based hesitation. The moderation hypothesis was not rejected but was also not confirmed, indicating that trust and risk operate as independent additive predictors in this sample. Random Forest validation confirmed the robustness of

these findings, with Perceived Usefulness and Trust ranking first and second in feature importance across both methods.

Consumer segmentation identified three distinct profiles: Enthusiasts (62%), who show high adoption willingness and AI familiarity; Pragmatists (14%), who prefer hybrid models retaining human control; and Sceptics (24%), defined by high perceived risk and low trust. Each segment requires a differentiated strategic approach: Enthusiasts respond to functional value demonstrations; Pragmatists require human-in-the-loop configurations that maintain meaningful oversight; and Sceptics need trust-building interventions through explainability, transparency, and gradual exposure to low-stakes AI applications. The study contributes an original empirical dataset, an extended TAM framework incorporating automation preference, and a combined analytical methodology applied to the underexplored Spanish consumer market.

1. Introduction

1.1 Background

The introduction of artificial intelligence (AI) has reshaped how consumers conduct tasks on the web and how they make purchase decisions. Advanced AI systems help users accomplish this by analyzing vast amounts of data, predicting user behavior, and assisting with complicated purchase decisions (Schumacher et al., 2026; Cuesta-Valiño et al., 2025). The UTAUT (Venkatesh et al., 2003) provides an accepted theoretical foundation for these technologies through three perceived dimensions: usefulness to the user, effort required by the user and social influence on the user.

AI shopping agents provide active support to shoppers during an entire buying process by collecting behavioral data from shoppers to compare alternative products and to create personalized suggestions based on each shopper's data in real time. In more advanced uses, they can also execute or complete purchases for the shopper, representing a major transition from purchase processes that are controlled by humans to purchase processes that are only partially or completely automated (Schumacher et al., 2026).

The primary reason people adopt AI technologies is because of the perceived benefit of having their shopping experience improved through AI's ability to reduce cognitive load associated with searching for, comparing and making purchasing decisions, thus increasing the quality of the overall decision-making and shopping experience (Cuesta-Valiño et al., 2025; Venkatesh et al., 2003). However, consumers are more hesitant to rely on automated systems for making decisions about subjective or high-involvement purchases than for routine, objectively based ones (Castelo et al., 2019); hence the extent to which the value of using an AI shopping agent is perceived will differ across different purchasing contexts and levels of personal involvement.

Despite these benefits, AI shopping agents also create significant challenges. By handing over control of decision-making to an automated system, the user may express concerns with regard to the visibility and accountability of the automation process (Schumacher et al., 2026); therefore, the effectiveness of automating shopping decisions will depend on the extent to which users' trust has been established through proper calibration (Lee & See, 2004). According to Shin (2021), trust in AI is also affected by the user's understanding of

how the AI system makes its decisions, and if the AI system is opaque in its reasoning, users will be less likely to delegate purchasing decisions to an AI system.

Deciding whether to adopt AI technology depends on two critical factors: trust and perceived risk. People are concerned about security, fraud, data privacy, and losing control when dealing with AI, as AI gains increasingly greater autonomy. A multi-dimensional risk framework (Featherman & Pavlou, 2003) found that financial, performance, psychological, social, time, and privacy risks were key barriers to adopting an e-service. Furthermore, Kim et al. (2008) documented the complementary relationship between trust and perceived risk, finding that a person's increase in trust directly decreases the impact of perceived risks when they intend to purchase a good or service; this finding is equally relevant to transactions with the use of AI systems. This relationship is particularly relevant for consumers in Spain, where there has been much use of AI tools among consumers, but consumers are still hesitant to rely solely on automated systems to make significant decisions due to concerns about data privacy and algorithmic transparency (Bain & Company, 2025; Iglesias, 2025; Euronews, 2025).

1.2 Challenge Overview and Knowledge Gap

Despite widespread recognition of AI's functional benefits, adoption of fully autonomous AI systems remains limited. In a study conducted by Bain and Company (2025), although a large number of consumers have tried using AI as a tool to locate and compare products, there are only a small number of consumers who are willing to use fully autonomous AI to make purchasing decisions. In fact, in a survey of over 2,000 consumers, only 24 percent stated that they are comfortable using autonomous AI for making purchasing decisions because of concerns related to privacy and security.

The primary barrier is the disparity between high levels of consumption and low levels of trust (Bain and Company, 2025; Riskified, 2025). While AI is recognised as a valuable tool for supporting decision-making, the level of trust in AI as a replacement for human decision makers remains a major barrier to adoption. Kim et al. (2008) showed that trust and perceived risk complement each other: when trust grows, it directly diminishes the negative impact of perceived risk on purchase intention. Furthermore, the involvement of an autonomous AI system as the decision-maker creates a new complexity within this relationship.

Most current research focuses exclusively on traditional platforms or recommendation systems, leaving autonomous AI shopping agents largely unexplored (Schumacher et al.,

2026). Castelo et al. (2019) found that customers exhibit a higher level of aversion to algorithm-based decision-making when purchasing in situations where customer involvement is high or subjective than when it is low or objective.

A related gap concerns automation and explainability trust. Trust has a positive influence on purchasing behaviour (Ibadi & Supriyono, 2025) and acts as a mediator between technology's benefits and users' behavioural intentions (Qalati et al., 2021); it is also a key driver of willingness to rely on automated systems (Lee & See, 2004). Shin (2021) showed that AI explainability significantly increases perceived trust and acceptance, yet explainability remains under-researched as most existing AI shopping agent implementations provide little transparency about their decision-making processes.

Another insufficiently researched area is perceived risk. Perceived risk has been shown to be a major impediment to adoption (Featherman & Pavlou, 2003), with empirical evidence confirming its negative effect on purchasing behaviour (Abarna et al., 2023). Dogbe et al. (2019) demonstrated that financial risk has the greatest negative impact on purchasing decisions, with trust moderating but not completely eliminating these effects. Rasty et al. (2021) identified privacy risk as the greatest obstacle to consumer trust, concerns that are further heightened by AI's introduction of new fraud, data misuse, and loss of control issues (León, 2025; Silverfort, 2025).

Furthermore, functional advantages alone cannot fully explain the adoption decision process. The relationship between e-service quality and consumer trust is mediated by perceived usefulness (Bhat & Darzi, 2020; Qalati et al., 2021). While the UTAUT framework (Venkatesh et al., 2003) provides a comprehensive lens for analysing these interrelated factors, it needs to be adapted for the autonomous nature of AI shopping agents, particularly regarding the delegation of decision-making authority.

This study addresses these gaps by developing an extended TAM framework that includes automation preference, and combining MLR, Random Forest, and K-means Clustering analyses in the underresearched Spanish consumer context.

1.3 Research Problem

The adoption of completely autonomous shopping agents is still uneven, despite the quick integration of AI into digital commerce. Seventy-three percent of consumers use AI to generate product ideas, summarize product reviews, and provide pricing comparisons (43%,

37%, and 32%, respectively). However, only 13% of consumers have made a purchase after an AI shopping assistant instructed them to do so (Riskified, 2025). As a result, there is a significant gap between usage and adoption since many customers use AI but do not completely trust or rely on it.

The biggest barrier to switching decision-making authority from the consumer to the AI shopping assistant is the psychological discomfort of transferring that level of power to an AI. While 7 out of 10 consumers feel somewhat comfortable when making an AI-assisted purchase, their main concerns are around payment security (32%), privacy (26%), possible errors (18%), and loss of control (17%), illustrating that these consumers have complicated and conflicting feelings of comfort and true confidence in the AI shopping assistant (Riskified, 2025; Bain & Company, 2025). A critical component of building trust in AI environments is that, unlike human vendors, AI shopping agents cannot rely on interpersonal relationships to generate confidence, making transparency and explainability essential substitutes.

This study examines how trust (Kim et al., 2008), perceived risk (Featherman & Pavlou, 2003), perceived usefulness (Davis, 1989; Venkatesh et al., 2003), automation preference (Lee & See, 2004), and adoption intention collectively influence consumers' willingness to adopt AI shopping agents. All five constructs have been tested individually in previous research but no studies have included all five as they relate to the adoption of AI shopping agents within a single unified model. This study fills that gap through an innovative empirical analysis of data obtained from a survey of Spanish consumers, an underexplored area of consumer research.

1.4 Research Objectives

The main objective of this research is to evaluate the factors influencing consumers' willingness to adopt AI-based purchasing agents in relation to business intelligence and data analytics. As organisations establish AI-powered purchasing solutions, it is critical to understand how human behaviour and psychology impact the decision-making process since AI is becoming more accepted in everyday digital commerce. The need to understand the foundations of adoption behaviour based on data is a growing priority for companies, given that agentic commerce continues to develop on a global scale and is projected to conduct anywhere from \$3-\$5 trillion worth of transactions by 2030 (Schumacher et al., 2026).

First, this research examines the level of consumer trust in AI-based systems and its influence on users' willingness to adopt AI shopping agents. There is clear evidence from studies conducted in the US market that consumers have a high degree of caution when purchasing products from an autonomous AI system; however, there is also a body of evidence from studies conducted in Spain that shows similar patterns of uncertainty regarding the delegation of decision-making authority to AI technologies. Iglesias (2025) documented that consumers show considerable uncertainty and difficulty using systems that provide high levels of autonomy in making purchasing decisions.

Second, the study examines how consumers perceive risk and how risk impacts their intention to use AI shopping agents. Perceived risk is particularly important in the context of AI shopping agents, as the biggest obstacle to gaining consumer trust is related to privacy concerns (Rasty et al., 2021). Moreover, the introduction of newer forms of security risk associated with autonomous systems, including data misuse and system vulnerabilities, creates additional reasons for consumers to be reluctant to trust an autonomous system (Silverfort, 2025).

Third, the study evaluates the impact of perceived usefulness on adoption behaviour. Within the realm of AI shopping agents, perceived usefulness refers to the system's ability to save search effort, make product evaluations easier, and enable consumers to make better purchasing decisions. Importantly, perceived usefulness also serves as a mediator between e-service quality and consumer trust, showing its importance both as an adoption facilitator and a trust-building factor (Bhat & Darzi, 2020).

Fourth, the study examines user preferences for automation and distinguishes between AI as a fully autonomous decision maker versus AI as a decision-support tool. This distinction is important because consumers are generally willing to use AI to assist them in making decisions, but they have less willingness to give AI full control. According to Lee & See (2004), appropriate reliance on automation, calibrated to the risk and complexity of the decision, is critical to effective human-automation interaction and therefore central to understanding AI shopping agent adoption.

Fifth, the study investigates how users behave differently depending on their demographic characteristics, such as age, and their experience using digital devices and AI tools. Evidence from Spain indicates that despite growing familiarity with AI tools, a large percentage of users remain reluctant to entrust significant decisions to AI systems (Iglesias, 2025),

highlighting that adoption is highly influenced by cultural and also by the demographic characteristics.

Finally, this research aims to develop a data-driven understanding of users' likelihood to adopt AI shopping agents, including regression analysis validated through Random Forest and user segmentation via K-means Clustering. From this, the research identifies the key characteristics of users who adopt AI and estimates their likely adoption behaviour across different consumer segments.

2. Theory

2.1 Literature Review

Examining important theoretical foundations on technology adoption, automation trust, and perceived risk is essential to comprehending users' willingness to adopt AI shopping agents. These concepts have been extensively researched in relation to digital consumer behaviour and e-commerce, offering a solid theoretical framework for examining how people engage with intelligent decision-support systems.

From the perspective of technology adoption, prior research indicates that users are more likely to accept digital systems when they perceive them as advantageous and capable of improving their performance (Davis, 1989; Venkatesh et al., 2003). The UTAUT paradigm specifically emphasizes perceived usefulness, effort expectancy, and social influence as important factors that drive adoption intention. When applied to autonomous AI systems, this framework must be modified due to the additional dimension of decision delegation involved. Unlike traditional technologies, AI shopping agents necessitate that users rely on the system for both information and decision-making; this level of autonomy considerably alters the adoption dynamic (Lee & See, 2004).

In this context, adoption is heavily influenced by trust. Trust lowers uncertainty and enhances consumers' willingness to interact with technology systems and make transactions online (Kim et al., 2008; Ibadi & Supriyono, 2025). Empirical studies demonstrate that trust mediates the relationship between behavioural intention and technological attributes such as reputation, website quality, and service quality (Qalati et al., 2021). Furthermore, Shin (2021)

showed that the explainability of AI systems greatly increases perceived acceptance and trust, indicating that transparency in AI decision-making is a crucial adoption enabler.

However, one major obstacle to adoption is perceived risk. Consumers' desire to accept digital technologies is adversely affected by financial risk, product ambiguity, data privacy concerns, and potential fraud (Featherman & Pavlou, 2003; Abarna et al., 2023). Dogbe et al. (2019) showed that financial risk has the strongest negative impact on mobile shopping behaviour, while Rasty et al. (2021) found that privacy risk is the single most important obstacle to consumer trust in online environments. These risks are amplified in AI-mediated environments by concerns about data misuse, system security, and opaque automated decision-making procedures (Silverfort, 2025; León, 2025).

Therefore, technological advantages alone cannot account for the uptake of AI shopping agents. Rather, adoption is determined by how consumers' desire to provide autonomous systems with decision-making authority interacts with perceived usefulness, trust, and perceived risk. Crucially, this willingness differs according to the type and difficulty of the purchase task: consumers are substantially more resistant to algorithmic decision-making in high-involvement or subjective purchase contexts than in routine ones (Castelo et al., 2019).

2.2 Technology Adoption and the Technology Acceptance Model (TAM)

The adoption of new technologies has been studied by a number of theoretical frameworks, the most well-known of which is Davis' (1989) Technology Acceptance Model. The Technology Acceptance Model seeks to explain how people will embrace and utilize new technology based on two crucial elements: perceived usefulness and perceived ease of use. Venkatesh et al. (2003) developed the Unified Theory of Acceptance and Use of Technology, which incorporates effort expectancy, social influence, and facilitating conditions as predictor variables into the Technology Acceptance Model.

Perceived usefulness reflects an individual's belief that they can use the technology to achieve better performance or make better decisions. For example, consumers may use AI shopping assistants because they perceive them as faster and more efficient than traditional search methods. In addition to being used as a driving force for the adoption of technology, perceived usefulness has also been shown to mediate the relationship between e-service quality dimensions and consumer trust, suggesting that perceived usefulness can also act as a trust-building mechanism (Bhat & Darzi, 2020). AI systems can increasingly be thought of as

interfaces to assist consumers in researching products and helping them make comparisons, and will continue to outperform traditional methods in doing so (Schumacher et al., 2026).

Yet, the original TAM model has limitations regarding the incorporation of autonomous technology like AI-based shopping agents. The TAM base assumption is that the human as the end user retains total control over his/her relationship with the technology, while relying on AI-based shopping agents to make decisions introduces varying degrees of delegation and autonomy. According to Lee & See (2004), this continuum from decision-support to full delegation corresponds to the continuum of calibrated trust needed for the use of AI-based shopping agents. Additionally, Castelo et al. (2019) identified that resistance to algorithmic decision-making varies based upon the context of the decision being made, indicating that as consumers are making more subjective or high-involvement purchases they have a greater degree of resistance.

In order to enhance the Technology Acceptance Model (TAM), it is necessary to include additional factors: trust and perceived risk. Under conditions of uncertainty, trust plays a key role in technology adoption (Kim et al., 2008; Ibadi & Supriyono, 2025), and it has been shown that an explainable AI system increases trust and leads to its acceptance (Shin, 2021). Conversely, perceived risk from potential financial loss, privacy concerns, and system errors negatively impacts the intention to adopt a technology. In particular, privacy-related risks are the primary barrier to developing consumer trust in online transactions and e-commerce sites (Featherman & Pavlou, 2003; Rasty et al., 2021). Consumers who use AI tools (73%) still do not fully rely on autonomous systems because of concerns regarding their level of trust in such systems, and users are also concerned with the system's transparency and security level when making this type of decision (Riskified, 2025; Bain & Company, 2025). Therefore, this study uses the extended TAM model of trust and perceived risk as critical drivers of behaviour along with perceived usefulness.

2.3 Trust in Automation and Autonomous Decision-Making Systems

Consumers' propensity to rely on automated systems is largely determined by their level of trust, especially in situations of high ambiguity and low transparency. According to Lee and See (2004), trust in automation is the conviction that a technology system will enable users to accomplish their objectives in vulnerable situations. Users are more likely to rely on a system when they believe it to be dependable, predictable, and capable of producing correct results.

The emergence of AI shopping agents fundamentally changes how consumers engage with technology. Trust is a key factor in determining adoption because AI agents may analyse alternatives, make recommendations, and even make purchases, unlike traditional systems that provide users with information for decision-making (Schumacher et al., 2026).

According to empirical research, trust mediates the relationship between technological attributes such as service quality and reputation and behavioural intention (Qalati et al., 2021), and has a strong and significant positive effect on purchase intention across various e-commerce contexts (Ibadi & Supriyono, 2025). Kim et al. (2008) also showed that trust and perceived risk are complementary forces: when trust grows, perceived risk's constraining influence on purchase intention is immediately diminished, a relationship especially pertinent in AI environments where uncertainty is naturally high. This is further supported by recent empirical data: Mani et al. (2025) showed that AI-based personalisation significantly increases consumer trust ($\beta = 0.68$), confirming its role as both an adoption driver and an outcome of positive AI interactions, while Luo and Zhang (2025) identified trust as the primary driver of AI agent acceptance across 632 participants.

Trust becomes considerably more complex in AI-driven systems. Shin (2021) showed that perceived trust and acceptance are greatly increased by AI explainability, the extent to which users can comprehend how and why a decision was made. Regardless of perceived efficiency, users are less inclined to assign decision-making authority to AI shopping agents that function as opaque black boxes. Consumers' main concerns include payment security, privacy, potential errors, and loss of control (Riskified, 2025; Bain & Company, 2025).

One strategy to close this gap is the adoption of human-in-the-loop systems, in which users retain some degree of control over automated decisions. Such hybrid models are especially pertinent given that consumer resistance to algorithmic decision-making varies greatly across buying contexts, reducing ambiguity and increasing confidence (Castelo et al., 2019; Schumacher et al., 2026). Additionally, Dogbe et al. (2019) demonstrated that trust functions as a substantial moderator between perceived risk and adoption behaviour, attenuating but not completely removing the negative impacts of risk.

2.4 Perceived Risk and Fraud Concerns in AI-Based Commerce

Perceived risk is widely considered one of the main determinants of technology acceptance, especially in digital and internet-based conditions. Featherman and Pavlou (2003) defined

perceived risk as the likelihood that one will incur a loss when making a choice in an unpredictable environment, identifying six key dimensions: financial, product, privacy, psychological, social, and time risks.

Perceived risks reduce the likelihood of online purchasing (Abarna et al., 2023). Rasty et al. (2021) identified privacy risk as the primary barrier to consumer trust in e-commerce environments, while Dogbe et al. (2019) showed that financial risk has the greatest negative impact on mobile shopping behaviour.

The use of AI shopping agents compounds existing risk concerns. Unlike traditional online shopping where the user retains more control, an AI agent evaluating and acting on the user's behalf introduces new uncertainty and vulnerability (Schumacher et al., 2026). Furthermore, Castelo et al. (2019) indicated that consumers are more resistant to algorithmic decision-making for highly involved or subjective purchases, highlighting that perceived risk is situationally specific and varies significantly depending on the nature and complexity of the purchase task.

In AI-mediated commerce, fraud and data exposure represent some of the highest-risk dimensions. According to survey data from Riskified (2025), payment security (32%) and privacy (26%) are the two primary concerns when consumers contemplate an AI-mediated purchase. Industry reports confirm that AI agents may increase risk through susceptibility to cyber attacks and misuse of sensitive information (Silverfort, 2025). Furthermore, incorporating AI into commercial transactions introduces new uncertainties around responsibility and liability when AI-assisted decisions are executed without direct user intervention (León, 2025).

The potential loss of control over decision-making is another major risk dimension. According to Lee and See (2004), a user's willingness to delegate control is influenced by their perception of system reliability and the value at stake. When users can see some degree of oversight over AI-assisted decisions, such as approval prior to execution, they are more likely to accept AI technologies (Schumacher et al., 2026).

Finally, perceived risk and trust are closely interrelated: as trust grows, the constraining influence of perceived risk on purchase intention diminishes, though trust moderates but does not completely eliminate risk-driven adoption barriers (Kim et al., 2008; Dogbe et al., 2019).

2.5 Artificial Intelligence and Consumer Decision-Making

In digital commerce settings, AI is transforming how consumers make decisions by streamlining the purchasing process, analyzing data, finding alternatives, and facilitating decision assessment (Schumacher et al., 2026; Cuesta-Valiño et al., 2025). AI-driven systems enable personalised recommendations based on user preferences and previous interactions, increasing consumer engagement and improving the shopping experience.

However, customer acceptability of high-autonomy AI shopping assistants varies greatly depending on the circumstances of the transaction, and adoption is largely moderated by perceived benefits (Frank & Otterbring, 2026).

Adoption of AI cannot be entirely explained by technological performance alone. If consumers believe AI-based systems are dependable, beneficial, and low-risk, they are more likely to adopt them (Ibadi & Supriyono, 2025; Abarna et al., 2023). Adoption of AI is not always hampered by psychological factors; in subjective or highly involved buying circumstances, resistance is far greater than in routine ones.

In summary, AI is changing consumer decision-making by increasing efficiency and facilitating personalisation; however, its adoption depends on the interplay of perceived usefulness, perceived risk, trust, and automation preferences, the fundamental variables of the conceptual model developed in this study.

2.6 Conceptual Model and Hypotheses

Building on the theoretical framework established in the preceding sections, this study proposes a conceptual model based on the extended TAM (Davis, 1989; Venkatesh et al., 2003), incorporating perceived usefulness, trust, perceived risk, and automation preference as additional determinants of adoption. In this model, perceived usefulness represents users' perception that AI shopping agents can support purchasing decisions more efficiently and improve decision-making effectiveness by acting as both an impetus to adopt and a vehicle for building trust (Bhat & Darzi, 2020). Trust is included because AI systems require users to rely on them for both information and decision-making, making trust a key driver of adoption under uncertainty (Kim et al., 2008; Ibadi & Supriyono, 2025; Shin, 2021; Qalati et al., 2021). Perceived risk, across financial, privacy, psychological, and other dimensions (Featherman & Pavlou, 2003), is expected to negatively influence adoption. Automation preference captures consumers' varying readiness to delegate decision-making; while

appropriate reliance on automation is essential for effective human-automation interaction (Lee & See, 2004), users remain more reluctant to embrace fully autonomous systems than AI as a decision-support tool (Schumacher et al., 2026; Riskified, 2025). Demographic and behavioural control variables; age, digital experience, and familiarity with AI, have been added because research suggests that these factors affect the relationship between user segments and adoption behaviour (Iglesias, 2025). Trust and perceived risk also interact significantly such that when individuals have high trust, they will experience a less negative effect from perceived risk on their intentions to adopt than someone who has low trust; this makes trust an important psychological moderator and therefore, an important factor to consider in this research study (Kim et al., 2008; Dogbe et al., 2019).

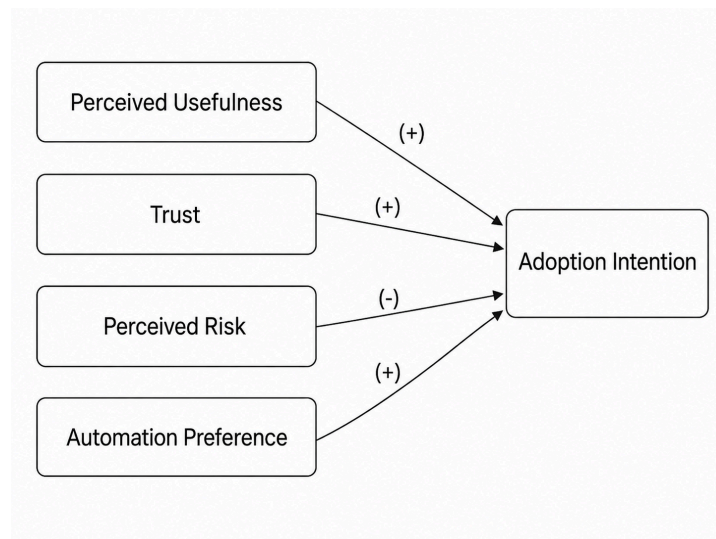


Figure 1. Conceptual Model of AI Shopping Agent Adoption

Hypotheses

It has been found by many researchers that perceived usefulness is one of the strongest predictors of technology adoption (Davis, 1989; Venkatesh et al., 2003), and is also a reflection of users' beliefs about whether AI shopping agents will provide them with improved efficiencies, and better decision-making capabilities when shopping. Supporting this idea, empirical research indicates that the perceived usefulness of e-services serves as a mediator between e-service quality and trust, thereby demonstrating its dual role as a driver of adoption, as well as a trust building mechanism (Bhat & Darzi, 2020; Schumacher et al., 2026).

H1: *Perceived usefulness has a positive effect on users' willingness to adopt AI shopping agents.*

Trust reduces uncertainty and enables reliance on digital systems; thus, it is a vital factor for AI systems, where the user has delegate their decision-making authority to the autonomous system (Kim et al., 2008; Ibadi & Supriyono, 2025). AI explainability increases the perception of trust (Shin, 2021), and trust mediates the relationship between technology attributes and behavioural intentions (Qalati et al., 2021). Trust is defined as a key driver of acceptance of AI agents (Luo & Zhang, 2025), and AI-personalisation increases consumer trust (Mani et al., 2025). Nonetheless, only 24% of consumers are willing to make an autonomous AI-based purchase, confirming that trust remains the primary adoption barrier (Bain & Company, 2025).

H2: *Trust in AI shopping agents has a positive effect on users' willingness to adopt them.*

The potential danger perceived by the consumer, which consists of elements such as financial losses, concern for privacy, fraud or loss of control, has been shown repeatedly to have a negative influence on intention to adopt digital commerce (Featherman & Pavlou, 2003; Abarna et al., 2023). According to Dogbe et al. (2019), the most significant risk involved with making a purchase is financial, while Rasty et al. (2021) indicated that privacy concerns are a major barrier to building customer trust. In AI-enhanced environments, these risks are heightened due to greater opportunities for security breaches and poor transparency associated with automated decision-making processes (Silverfort, 2025).

H3: *Perceived risk has a negative effect on users' willingness to adopt AI shopping agents.*

Differences exist between consumers who are willing to have an automated system make decisions on their behalf; therefore, a consumer who is comfortable using an automated system is much more likely to accept using autonomous systems (Lee & See, 2004), whereas most consumers prefer hybrid models, where AI tend to assist them and a human is in control (Schumacher et al., 2026; Riskified, 2025). Aversion to algorithms is task-dependent and differs based on the context of the purchase (Castelo et al., 2019); therefore, acceptance of

highly autonomous AI systems depends on consumers' perceived benefits and comfort with delegation (Frank & Otterbring, 2026).

H4: *Users' preference for higher levels of automation positively influences their willingness to adopt AI shopping agents.*

Previous studies have shown that both perceived risk and trust function as complementary and interdependent elements; as trust increases, perceived risk's impact on purchase intentions decreases (Kim et al., 2008). Prior empirical evidence supports the idea of a moderating relationship between trust and risk-based adoption barriers in mobile and digital commerce settings. However, while trust has the potential to reduce risk-based adoption barriers, it does not always eliminate them completely (Dogbe et al., 2019).

H5: *Trust moderates the negative effect of perceived risk on users' willingness to adopt AI shopping agents.*

3. Methodology

3.1 Research Design

This study adopts a quantitative research design to examine the factors influencing consumer adoption of AI-based shopping agents. A quantitative approach is appropriate because it enables the systematic measurement of consumer perceptions, attitudes, and behavioural intentions through structured data, and facilitates the empirical testing of theoretically grounded hypotheses, specifically H1 through H5 derived from the extended TAM framework. Unlike qualitative methods, which prioritise depth of interpretation over generalisability, a quantitative design allows for statistical inference across a sample and supports the identification of directional relationships between constructs such as trust, perceived risk, perceived usefulness, and automation preference.

The study utilises a cross-sectional design, where data are gathered at a single point in time, to gain insights about how consumers understand AI shopping agents and where they are still developing their overall attitudes toward this rapidly evolving technology. This design is well-suited for capturing a snapshot of current consumer perceptions across different user profiles.

A Business Intelligence and Data Analytics framework is employed because this study goes beyond descriptive analysis to generate actionable, data-driven insights into consumer adoption behaviour. BI and DA approaches are particularly suited to this context as they combine hypothesis testing with predictive modelling and consumer segmentation, enabling both explanatory and prescriptive outputs. This is consistent with the study's dual academic and managerial objectives: to contribute to the theoretical understanding of AI adoption, and to provide businesses with empirically grounded strategic guidance for AI-driven commerce environments.

Dimension	Description
Approach	Quantitative
Design	Cross-sectional survey
Framework	Business Intelligence & Data Analytics (BI/DA)
Sample	N = 200 valid responses
Instrument	Online structured survey (Google Forms)
Sampling method	Non-probability convenience sampling
Scale	5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree)
Analysis methods	Cronbach's Alpha, Pearson Correlation, MLR, Random Forest, K-means Clustering

Table 1. summarises the key dimensions of the research design.

The analytical framework of this research consists of five sequential procedures: Reliability Analysis (Cronbach's Alpha), Bivariate Correlation Analysis (Pearson), Multiple Linear Regression (MLR), Non-linear Validation (Random Forest), and Consumer Segmentation (K-Means Clustering).

3.2 Data Collection

Data were collected through an online survey administered using Google Forms. Google Forms was selected for its accessibility, ease of distribution, and ability to generate structured data suitable for subsequent statistical analysis.

The survey was distributed through electronic channels including social networks such as Instagram and LinkedIn, personal acquaintances, academic connections, and professional networks. This multi-channel approach was designed to reach a diverse range of respondents with varying levels of familiarity with online shopping and AI tools.

The survey was sent to approximately 350 individuals across these channels. A total of 200 valid and complete responses were collected, representing a response rate of approximately 57%. All responses were fully completed and retained for analysis, with no records excluded due to missing values.

Because participants were chosen based on their accessibility and willingness to participate, a non-probability convenience sampling technique was employed. This sampling strategy is appropriate for exploratory empirical research and for identifying relationships, behavioural patterns, and adoption dynamics in a growing field, even though it restricts the generalisability of findings to broader populations.

The dataset and analytical code used in this study are publicly available at:

<https://github.com/alisonppppppp/TFG-AI-Shopping-Agents>

3.3 Survey Design

This study aimed to gather data that provided insight into respondents' demographics, digital habits, familiarity with AI technology, and perceptions of AI-assisted shopping agents. The survey consisted of a series of sections that were administered using structured question formats.

Demographic Data - The first portion of the questionnaire asked respondents to provide demographic data in order to categorize them based on various relevant characteristics such as age, gender, country, and education level. By gathering this information, the results could be interpreted within context and cross-national differences could be compared among different groups of consumers.

Behavioural Data - The second portion of the questionnaire focused on establishing respondents' online shopping behaviours. Information relating to how often respondents made purchases online, as well as their history of using AI systems/technologies, helped to determine the extent to which respondents had been exposed to digital commerce systems.

AI-Related Questions - The third part of the survey asked respondents questions about AI shopping agents in order to provide information about their perceptions regarding using AI technologies while making purchases. Most of the attitude measures were assessed using a 5-point Likert scale ranging from 1 = strongly disagree to 5 = strongly agree. This was done because it allows respondents to express an opinion with varying degrees of agreement to a given statement, and it is widely used in quantitative research.

The questionnaire was designed to be concise and easy to complete in order to reduce respondent fatigue and improve response quality. The measurement items, variables, and example questions are presented in the Appendix.

3.4 Variables and Measurement

This study builds and assesses a research framework to investigate the potential determinants of consumer adoption of AI shopping agents. Both primary and secondary sources of literature are used to construct a model for the dependent variable (Adoption Intention) as well as the independent variables.

Adoption intention represents consumers' willingness to use AI shopping agents in future purchasing decisions.

The four primary independent variables are defined as follows:

Trust refers to the degree that a person trusts the potential of an AI shopping assistant, which includes the trustworthiness, reliability, dependability and responsiveness that the AI has to carry out consumer-related tasks in a way that maximises satisfaction. The influence of an AI system on the way a consumer makes purchasing decisions may completely or partially depend on the level of trust the consumer has in the AI (Kim et al., 2008).

Perceived risk refers to the respondent's level of concern about the potential harm from utilising an AI shopping agent, including concerns regarding privacy, financial loss, fraud, and lack of control or mistakes. Based on previous research, perceived risk is expected to have a negative impact on the likelihood of adopting AI shopping agents (Featherman & Pavlou, 2003).

Perceived usefulness refers to the extent to which respondents perceive that using AI shopping agents will lead to a more convenient, quicker, and more efficient shopping experience, thus aiding them in making better purchasing decisions. Perceived usefulness is

expected to positively impact the likelihood of adopting AI shopping agents (Venkatesh et al., 2003).

Automation preference refers to the level of comfort that the respondent has in delegating decision-making authority to an AI shopping agent. This variable is important to this study because AI shopping agents are fundamentally different from traditional recommendation systems by having higher levels of decision-making autonomy. Items AP1, AP2, and AP3 were reverse-coded prior to any analyses so that higher scores represented increased resistance to automation, consistent with the direction of H4.

Apart from the key explanatory variables, control variables were also examined. These included demographic variables (age, gender, education), digital experience, online shopping frequency, and familiarity with AI tools. The control variables helped identify how and why individuals behave differently when adopting a new product or service based on their personal characteristics and behaviours.

Table A5 in the Appendix provides a complete overview of all variables, their survey items, and their role in the analytical model, including the interaction term for H5 and the reverse coding applied to Automation Preference items.

3.5 Data Analysis Methods

Given the quantitative nature of the study, a five-step analytical framework was applied to the survey data. This framework was specifically designed to test hypotheses H1 through H5 in a rigorous and comprehensive manner. The five steps are both necessary and sufficient: Steps 1 and 2 establish the reliability and bivariate validity of the constructs before any inferential analysis is conducted; Step 3 directly tests all five hypotheses simultaneously through multivariate regression; Step 4 validates the robustness of those findings through a non-parametric method; and Step 5 extends the analysis to generate actionable consumer segmentation insights consistent with the study's Business Intelligence objectives.

Step	Method	Purpose	Key Output
Step 1	Cronbach's Alpha	Internal consistency of scales	α coefficient per construct
Step 2	Pearson Correlation	Bivariate relationships between variables	r matrix, p-values, direction

Step 3	Multiple Linear Regression (MLR)	Main hypothesis testing (H1–H5)	β , R^2 , F-stat, VIF, std. betas
Step 4	Random Forest	Non-linear validation of MLR	Feature importance, CV R^2
Step 5	K-means Clustering	User segmentation by profile	Cluster profiles, Silhouette score

Table 2. summarises the five analytical steps and their respective outputs.

Step 1 - Cronbach's Alpha. Internal consistency must be confirmed before any inferential analysis, as unreliable scales would undermine the validity of H1 through H5. Cronbach's Alpha was used to evaluate each multi-item scale's internal consistency, assessing whether items within a scale consistently reflect the same underlying construct, with $\alpha \geq 0.70$ regarded as the minimum acceptable threshold (Nunnally, 1978).

Step 2 - Pearson Correlation. Before the multivariate analysis, bivariate correlations offer preliminary directional evidence for every hypothesis. For each of the primary composite variables, Pearson correlation coefficients (r) were calculated, and significance was evaluated at $p < .05$, $p < .01$, and $p < .001$.

Step 3 - Multiple Linear Regression (MLR). This is the primary step, directly testing H1 through H5 simultaneously while controlling for all other predictors and accommodating the interaction term required by H5. To reduce multicollinearity, trust and perceived risk were mean-centred before calculating the interaction term. Standardised beta coefficients (std. β) are presented alongside unstandardised betas to enable direct comparison of predictor importance. R^2 , Adjusted R^2 , and the F-statistic were used to evaluate model fit. Every variance inflation factor (VIF) value was verified to be below the recommended cutoff of 5.

Step 4 - Random Forest Regression. As MLR assumes linearity, Random Forest provides non-parametric validation confirming whether the same variables drive adoption regardless of modelling approach. The model was trained on 80% of the data and evaluated on the remaining 20%, with 5-fold cross-validation applied to assess generalisability. Key hyperparameters were set as follows: $n_estimators = 500$, $min_samples_split = 5$, $min_samples_leaf = 2$, $random_state = 42$. Feature importance scores are reported and compared against the MLR standardised betas to assess consistency of findings across methods.

Step 5 - K-means Clustering. Hypothesis testing identifies which variables drive adoption but not how different consumers vary in their profiles; K-means clustering addresses this by generating actionable consumer typologies. Different consumer segments were identified using an unsupervised K-means clustering technique based on respondents' response profiles across the five primary composite variables. Before clustering, z-scores were used to standardise each variable. The Elbow Method and the Silhouette Score were used to calculate the ideal number of clusters. The following Python libraries were used for all analyses: pandas, numpy, scipy, scikit-learn, matplotlib, and seaborn.

All statistical analyses were performed within Python 3.11 using the following libraries: pandas and numpy for data manipulation, scipy for statistical testing, scikit-learn for Multiple Linear Regression, Random Forest, and K-means clustering, and matplotlib and seaborn for data visualization.

3.6 Ethical Considerations

Several ethical principles were upheld throughout the data collection process to ensure the integrity of the research and the protection of participants.

All survey participants were informed of the academic purpose of the study and provided implicit consent by completing the questionnaire, ensuring a transparent and informed participation process.

Responses were collected anonymously. No personally identifiable information beyond that which was strictly necessary for the analysis was collected. This helped to ensure that respondents remained private and were therefore more likely to provide truthful responses.

In addition to ensuring anonymity, the questionnaire was designed in such a way that it does not contain any misleading or excessively intrusive questions; this was particularly important for the current study because some of the topics related to trust, privacy, and the use of artificial intelligence are also issues related to the public's perception about how technology and their data are used by businesses.

4. Results

4.1. Sample description and descriptive statistics

A total of 200 valid and complete responses were collected. The sample is predominantly young, with 50.0% of respondents aged 18-24 and 34.5% aged 35-44. The sample consists of 56.5% male respondents, with 81.0% based in Spain. In terms of education, 65.0% of respondents have a bachelor's degree and 13.5% of the people have a master's degree. Respondents rated their AI familiarity on a 1-5 scale, with 36.0% rating themselves at level 4 and 23.0% at level 5, suggesting that the sample is generally above average in familiarity with AI tools. Figure 2 provides a summary of the key demographic distributions.

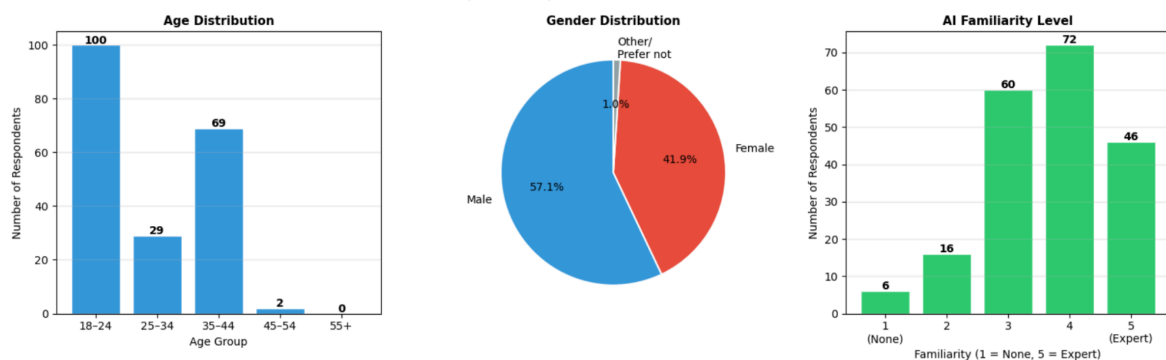


Figure 2. Sample Descriptive Statistics: Age, Gender, and AI Familiarity (N = 200)

The descriptive statistics for the five composite variables are presented in the Appendix (Table A2).

4.2 Reliability Analysis → Cronbach's Alpha (Step 1)

Internal consistency results are displayed in Table 2 and Figure 3.

The Trust construct was shown to have Excellent reliability ($\alpha = 0.909$). For the Perceived Usefulness and Adoption Intention constructs, Good reliability was obtained (both $\alpha = 0.852$), while Acceptable reliability was obtained for Perceived Risk ($\alpha = 0.749$).

In contrast, the Automation Preference construct fell below the reliability threshold ($\alpha = 0.671$); Questionable, with low item-total correlation, particularly for item AP3 (item-total $r = 0.430$). Despite this, it is retained because of its theoretical importance (Lee & See, 2004; Castelo et al., 2019), and this limitation has been noted.

Variable	Items	α	Interpretation	Item-Total r	α if Item Deleted (range)
Perceived Usefulness	PU1–PU 3	0.852	Good	0.693–0.773	0.751–0.821
Trust	TR1–TR 4	0.909	Excellent	0.762–0.825	0.871–0.894
Perceived Risk	PR1–PR4	0.749	Acceptable	0.493–0.591	0.663–0.718
Automation Preference	AP1–AP 3	0.671	Questionable	0.430–0.544	0.501–0.652
Adoption Intention	AI1–AI3	0.850	Good	0.709–0.737	0.775–0.801

Table 3. Reliability Analysis: Cronbach's Alpha by Construct

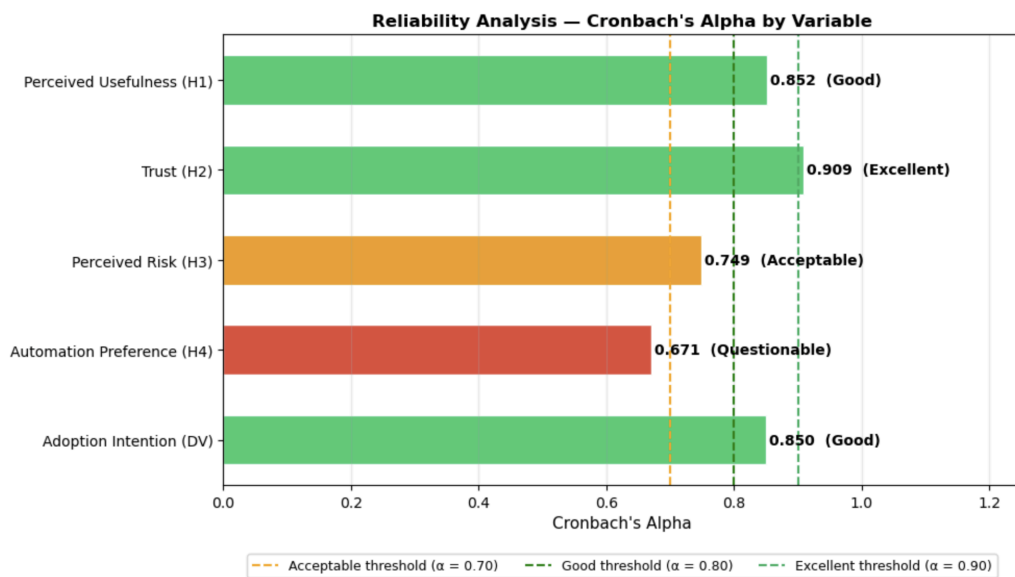


Figure 3. Cronbach's Alpha by Construct: Reliability Summary

4.3 Correlation Analysis → Pearson Correlation (Step 2)

The complete Pearson correlation matrix is provided in the Appendix (Table A3).

As shown in Figure 4, it can be seen that there are strong positive correlations between Adoption Intention and both Perceived Usefulness ($r = .714$; $p < .001$) and Trust ($r = .686$; $p < .001$), which provide preliminary support for Hypotheses 1 and 2. Perceived Risk ($r =$

-.146; $p = .040$) and Automation Preference ($r = -.160$; $p = .023$) display moderate but statistically significant negative correlations with Adoption Intention, giving directional indications of the relationships for Hypotheses 3 and 4.

Additionally, Perceived Usefulness and Trust are strongly positively correlated with one another ($r = .770$; $p < .001$), while Automation Preference and Perceived Risk are negatively correlated ($r = -.444$; $p < .001$). Trust and Perceived Risk presented a moderate negative correlation ($r = -.153$; $p = .031$), providing bivariate context for the moderation hypothesis H5. The heatmap indicates that in terms of the relationships associated with Adoption Intention, the strongest associations are to be found for Perceived Usefulness and Trust, whereas the bar chart helps illustrate how each independent variable is associated with the dependent variable by direction and strength.

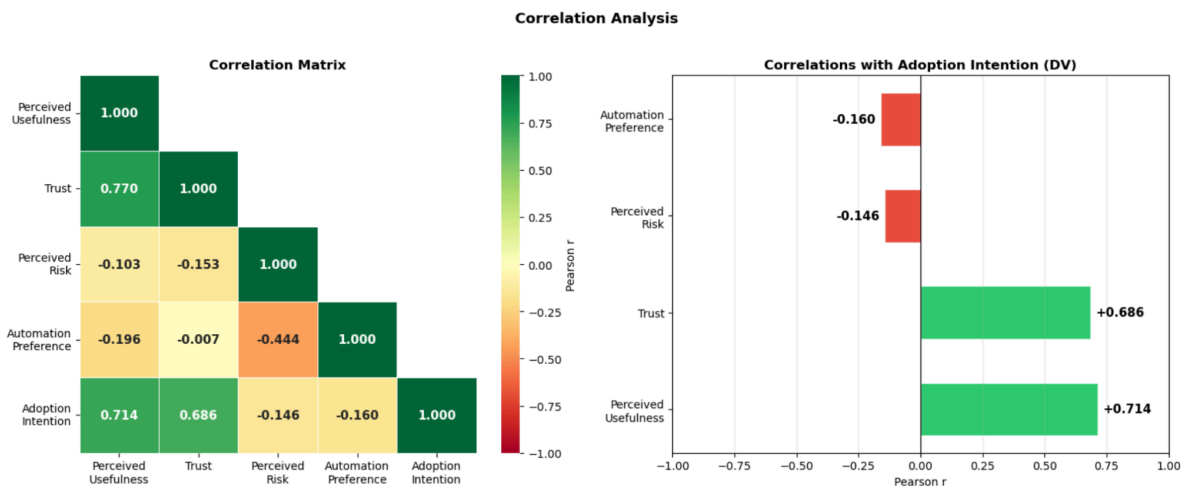


Figure 4. Pearson Correlation Matrix and Correlations with Adoption Intention

4.4 Multiple Linear Regression → Main Analysis (Step 3)

The main analysis was Multiple Linear Regression (MLR) with five predictors: Automation Preference, Perceived Usefulness, Trust, Perceived Risk, and the mean-centered interaction term $\text{Trust} \times \text{Perceived Risk}$. The dependent variable it was Adoption Intention. The whole model explained 57.1% of the variance in Adoption Intention ($R^2 = .571$, Adjusted $R^2 = .560$), and it was statistically significant ($F(5, 194) = 51.733$, $p < .001$). There were also no multicollinearity difficulties because all Variance Inflation Factor (VIF) values were less than 3.0. Table 4 and Figure 5 provide complete results.

H1 Perceived Usefulness ($\beta = +0.392$, $p < .001$):

The strongest predictor of Adoption Intention. Consumers are significantly more inclined to use AI shopping agents if they think they increase decision-making effectiveness and minimize search effort. This is consistent with TAM (Venkatesh et al., 2003; Davis, 1989). H1 is therefore not rejected.

H2 Trust ($\beta = +0.364$, $p < .001$):

Trust is the second strongest predictor. In line with earlier research showing that trust is a major factor in the adoption of AI (Ibadi & Supriyono, 2025; Luo & Zhang, 2025; Mani et al., 2025). H2 is therefore not rejected.

H3 Perceived Risk ($\beta = -0.099$, $p = .069$):

The multivariate model does not support H3 because the effect did not approach statistical significance at $p < .05$, despite the expected negative direction being seen. A weak trend is suggested by the marginal p-value and significant bivariate correlation ($r = -.146$, $p = .040$), which may be absorbed by the major impacts of perceived usefulness and trust. H3 is therefore rejected.

H4 Automation Preference ($\beta = -0.126$, $p = .024$):

There is a statistically significant effect, although it is in the other direction. This result shows that resistance to automation is linked to lower Adoption Intention, which is conceptually compatible with H4, since higher scores on this reverse-coded scale suggest greater resistance to automation. Among the important predictors, the effect is the least. H4 is partially not rejected.

H5 Trust $\times$ Perceived Risk ($\beta = +0.052$, $p = .272$):

The moderation hypothesis is rejected. Trust and Perceived Risk operate as independent additive predictors rather than in a multiplicative moderation relationship, contrary to the theoretical expectation of Kim et al. (2008) and Dogbe et al. (2019). H5 is rejected.

Predictor	B	SE	t	p	sig.	std. β	VIF
Perceived Usefulness	0.415	0.082	5.030	< .001	***	0.392	2.77
Trust	0.376	0.078	4.805	< .001	***	0.364	2.60

Perceived Risk	-0.117	0.064	-1.828	.069	ns	-0.099	1.32
Automation Preference	-0.159	0.070	-2.273	.024	*	-0.126	1.40
Trust × Perceived Risk	0.055	0.050	1.101	.272	ns	0.052	1.03

Note. Dependent variable: Adoption Intention. $R^2 = .571$, Adjusted $R^2 = .560$, $F(5, 194) = 51.733$, $p < .001$. All VIF values < 5 . * $p < .05$ *** $p < .001$.

Table 4. Multiple Linear Regression Results: Predictors of Adoption Intention

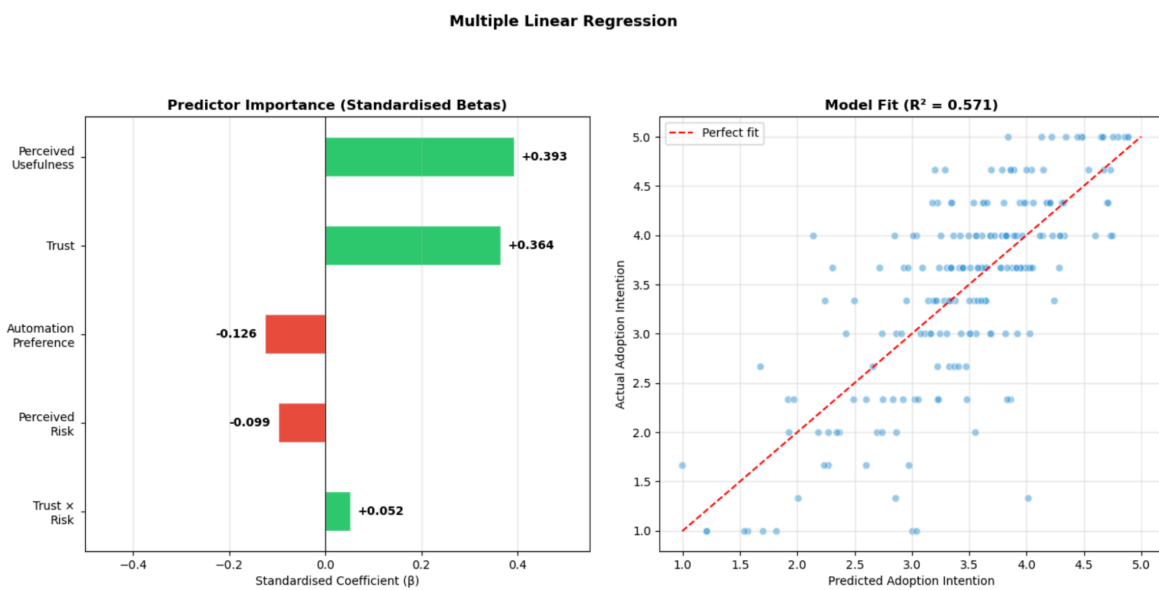


Figure 5. Multiple Linear Regression Results

4.5 Random Forest → Validation (Step 4)

A Random Forest Regressor (500 trees) was calculated on the same four independent variables to validate the MLR findings from a non-parametric, non-linear standpoint. To assess generalizability, an 80/20 train-test split and 5-fold cross-validation were applied.

The training R^2 was 0.853, indicating an excellent in-sample fit. The test R^2 was 0.174, and the cross-validated R^2 was 0.489 (SD = 0.067). The discrepancy between training and test R^2 indicates moderate overfitting with 200 observations. Cross-validated R^2 of 0.489 provides the most accurate indication of predictive performance. MAE = 0.619 and RMSE = 0.878 imply moderate prediction error on a scale of 1 to 5.

The complete feature importance comparison is provided in the Appendix (Table A4).

Additionally, the Random Forest feature importance scores converge with the MLR standardised betas in terms of variable ranking. Perceived Usefulness is ranked highest in all techniques (RF: 52.5%; MLR std. $\beta = 0.392$), followed by Trust (RF: 31.3%; MLR std. $\beta = 0.364$). The convergence across fundamentally distinct modelling methodologies adds robustness to the core discovery. Figures 6 and 7 show the results.

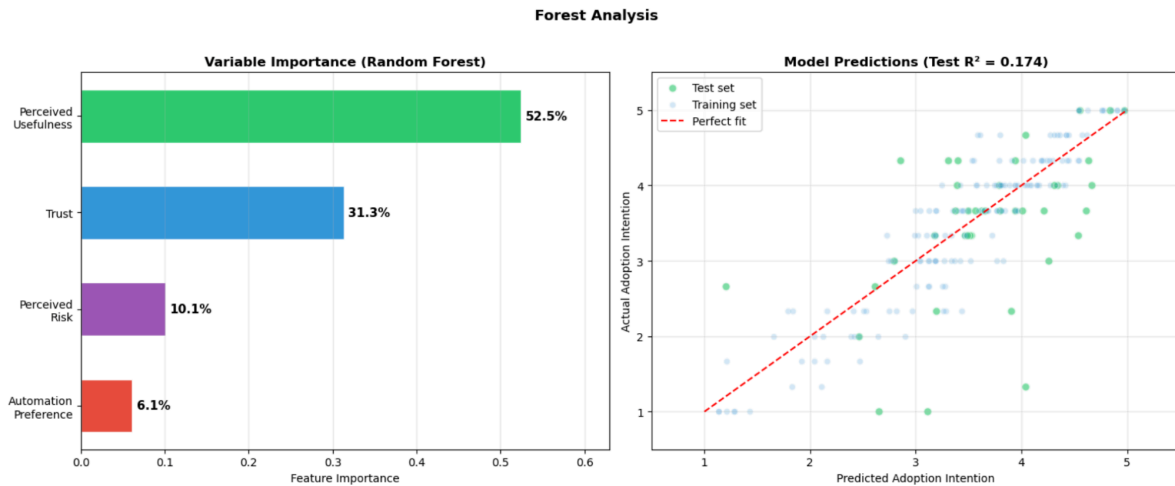


Figure 6. Random Forest: Feature Importance and Predicted vs. Actual Values

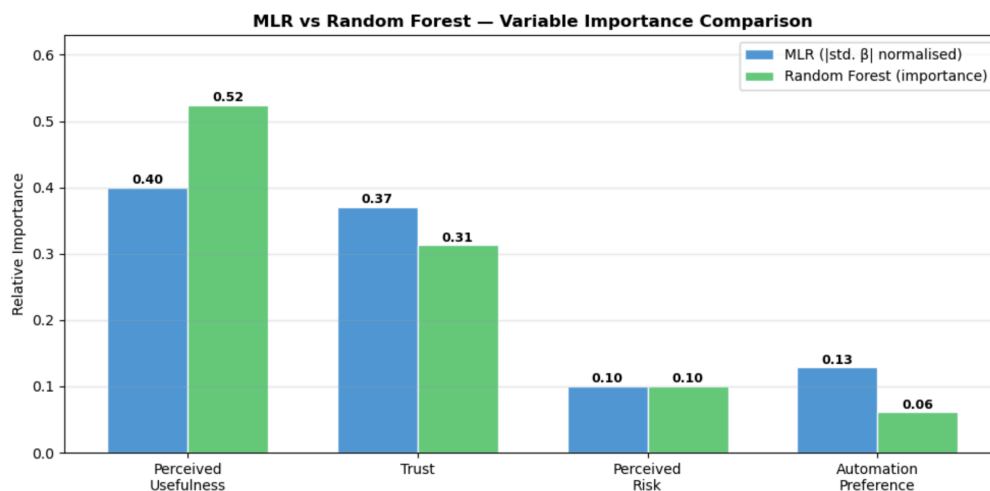


Figure 7. MLR vs. Random Forest: Variable Importance Comparison

4.6 K-Means Clustering → Consumer Segmentation (Step 5)

A K-means clustering technique was used to discover separate consumer categories based on five composite variables. Variables were standardized before grouping. The Elbow Method and Silhouette Score were assessed for K values ranging from 2 to 8 (Figure 8). The maximum silhouette score was at K = 2 (0.315), but K = 3 (0.308) was chosen for increased

theoretical and interpretative depth. Three meaningful profiles emerged, visualised in Figure 9. The complete cluster profiles are provided in the Appendix (Table A5).

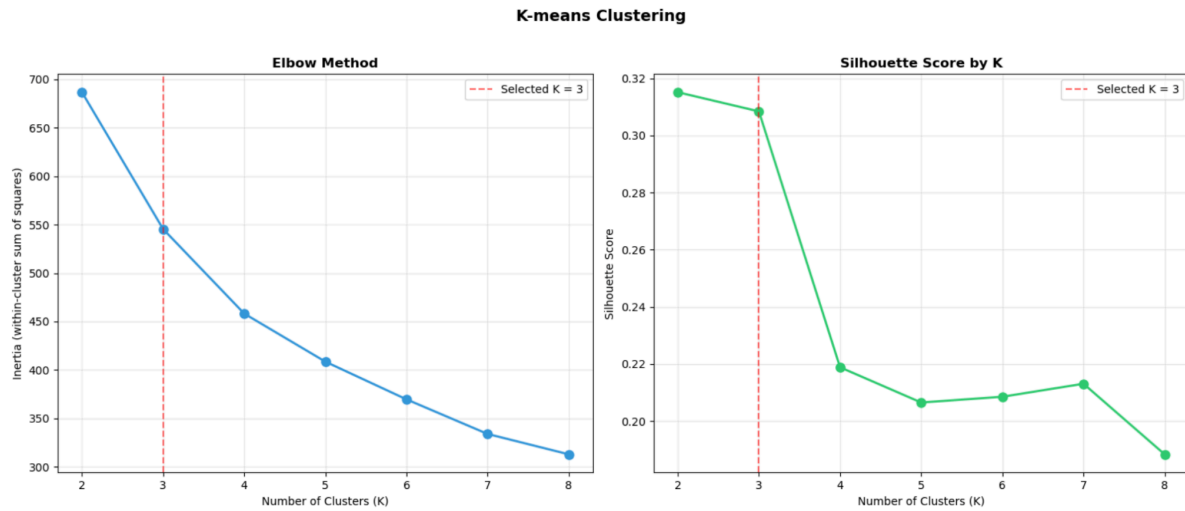


Figure 8. Elbow Method and Silhouette Score: Determining Optimal Number of Clusters

Enthusiasts ($n = 124$; 62%). The largest section has the highest Perceived Usefulness ($M = 4.056$), Trust ($M = 3.607$), and Adoption Intention ($M = 4.019$), along with moderate Perceived Risk ($M = 3.821$) and high automation comfort (AP $M = 1.457$). This segment is younger, more AI-familiar, and represents the primary market for AI shopping agent adoption.

Pragmatists ($n = 29$; 14%). The smallest group has intermediate scores on most variables (Adoption Intention $M = 3.092$, Trust $M = 3.052$) but significantly low Perceived Risk ($M = 2.655$) and the greatest resistance to automation (AP $M = 3.103$). Pragmatists are eager to interact with AI agents but strongly prefer to maintain control, which is consistent with hybrid model preferences (Schumacher et al., 2026).

Sceptics ($n = 47$; 24%). Defined by low Perceived Usefulness ($M = 2.560$), very low Trust ($M = 2.080$), high Perceived Risk ($M = 4.287$), and the lowest Adoption Intention ($M = 2.262$). This profile corresponds to the trust gap identified in industry research (Bain & Company, 2025; Euronews, 2025). Sceptics are typically older, less knowledgeable with AI, and require focused interventions to develop trust.

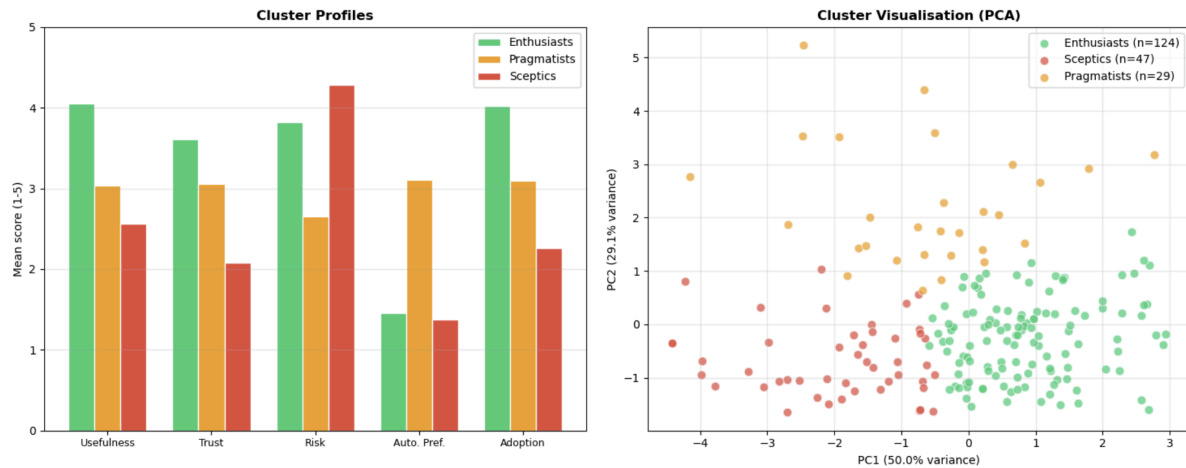


Figure 9. Cluster Profiles (bar chart) and Cluster Visualisation (PCA 2D)

4.7 Summary of Hypothesis Testing

Table 5 provides a consolidated overview of the five hypotheses, integrating findings from the correlation and regression analyses.

H	Hypothesis	Expected	Found	Result
H1	Perceived Usefulness → Adoption Intention (+)	Positive	$\beta = +0.392^{***}$	NOT REJECTED
H2	Trust → Adoption Intention (+)	Positive	$\beta = +0.364^{***}$	NOT REJECTED
H3	Perceived Risk → Adoption Intention (-)	Negative	$\beta = -0.099$, $p = .069$	REJECTED
H4	Automation Preference → Adoption Intention (+)	Positive	$\beta = -0.126^*$	REJECTED (reversed)
H5	Trust moderates Perceived Risk → Adoption (interaction)	Positive	$\beta = +0.052$, $p = .272$	REJECTED

Table 5. Summary of Hypothesis Testing Results

In conclusion, H1 and H2 are not rejected, confirming perceived usefulness and trust as the primary drivers of AI shopping agent adoption. H3 shows the expected negative direction but does not reach significance in the multivariate model and is therefore rejected. H4 is rejected in its expected direction but shows a statistically significant effect in the opposite direction, consistent with the reverse-coded scale. H5 is rejected, indicating that trust and risk operate as independent additive predictors in this sample.

5. Discussion and Conclusions

5.1 Discussion of Findings

H1- Perceived Usefulness (not rejected). Adoption intention was shown to be most strongly predicted by perceived usefulness (std. $\beta = 0.392$, $p < .001$). According to Davis (1989) and Venkatesh et al. (2003), consumers who view AI shopping agents as useful tools for decreasing search effort and enhancing decision quality are significantly more inclined to embrace them, which is fully consistent with TAM. The outcome also supports the dual role that Bhat & Darzi (2020) identified: perceived usefulness serves as both a direct adoption driver and a trust-building mechanism. This result is further supported by Schumacher et al. (2026), who show that the ability of AI agents to reduce cognitive effort in purchase is a major adoption factor.

H2 - Trust (not rejected). The second most powerful predictor was trust (std. $\beta = 0.364$, $p < .001$). Trust has been repeatedly found to be a major factor in the adoption of AI (Ibadi & Supriyono, 2025; Luo & Zhang, 2025; Mani et al., 2025). It also lessens ambiguity and mediates the relationship between technological qualities and behavioral intention (Qalati et al., 2021). Trust is a crucial psychological factor that makes up for the lack of human connection and the opaqueness of algorithmic decision-making as AI shopping agents transition toward autonomous transaction execution (Lee & See, 2004; Shin, 2021). Trust in AI transparency seems particularly important in the Spanish consumer setting, where data privacy worries are still prevalent (Iglesias, 2025).

H3 - Perceived Risk (rejected). Despite Perceived Risk's expected negative direction (std. $\beta = -0.099$), its effect was not statistically significant ($p = .069$). A suppression effect is a plausible explanation: once Perceived Usefulness and Trust are controlled for a large portion of the variance that would otherwise be attributed to risk is absorbed by their explanatory dominance. The substantial collinearity between Perceived Usefulness and Trust ($r = .770$) and the robust bivariate correlation between Trust and Adoption Intention ($r = .686$) confirm this. According to Frank & Otterbring (2026), who demonstrate that perceived benefits mediate the relationship between AI autonomy and adoption intention, consumers express moderate-to-high adoption willingness despite perceived risk being the sample's highest-scoring variable ($M = 3.761$). This suggests that usefulness perceptions can override risk-based hesitation.

H4 - Preference for Automation (partially rejected). A statistically significant negative effect (std. $\beta = -0.126$, $p = .024$) was generated by Automation Preference. This result is conceptually compatible with H4: respondents who oppose automation exhibit lower Adoption Intention since the scale was reverse-coded, with higher scores indicating greater opposition to automation. The majority of respondents tend against automation comfort, as seen by the sample mean being the lowest of all dimensions ($M = 1.677$). The hybrid-model choice shown by Schumacher et al. (2026) and Castelo et al. (2019) is reflected in the Pragmatist cluster (14%), which has the largest resistance to full automation (AP $M = 3.103$) despite low perceived risk.

H5 - Moderation of Trust x Perceived Risk (rejected). In contrast to Kim et al. (2008) and Dogbe et al. (2019), the moderation hypothesis was rejected (std. $\beta = 0.052$, $p = .272$). There are two possible explanations: first, there is little variance for the interaction term to explain because the independent effect of perceived risk is already minimal. Second, the contextual significance of the trust–risk interaction may be diminished by the sample's above-average AI familiarity because users who are at ease with AI might not rely on trust as a psychological buffer against risk in the same manner as less familiar customers.

5.2 Theoretical Contributions

This study makes three advances to the theoretical understanding of AI adoption in digital commerce. First, it expands on TAM by adding components like trust, perceived risk, and automation preference that are especially appropriate for autonomous AI systems, which are situations in which algorithms are given partial or complete control over decision-making. The current methodology operationalizes the distinct dynamics of consumer-facing AI agents, whereas earlier TAM extensions (Venkatesh et al., 2003) were created for organizational technology adoption.

Second, by empirically demonstrating that trust functions as an independent additive predictor rather than a pure mediator of risk in this situation, the study adds to the body of knowledge on trust. This improves on Kim et al.'s (2008) complementarity framework and implies that, in samples with above-average AI familiarity, perceived usefulness may take the lead in adoption, partially displacing the trust–risk dynamic.

Third, the integration of MLR, Random Forest, and K-means Clustering represents a methodologically innovative contribution. The segmentation analysis converts statistical connections into useful consumer categories, and the convergence of feature importance rankings across parametric and non-parametric models offers cross-validated evidence for the robustness of the primary findings.

5.3 Practical Contributions

The results have direct implications for companies that are developing and/or using AI-based shopping solutions, particularly in Spain and the European market more broadly. The empirical evidence provides three strategic recommendations:

Communicate usefulness rather than risk. The high level of perceived usefulness means that businesses should design their messaging around demonstrating concrete value such as time saved, improved decision quality, and personalised service. Regarding enthusiasts (62%), tangible demonstrations of functional benefits are more effective than generic privacy assurances in delivering utility-based value.

Build trust through transparency and explainability. Trust is the 2nd most significant predictor of acceptance of AI, and building trust is highly correlated with the degree to which a user understands how recommendations are developed through AI (Shin, 2021). Companies can promote built trust through designing interfaces for explainable AI so that users can see why a recommendation was made, displaying data-use policies, and providing users with the ability to choose to participate in the personalisation of their experience. This is particularly important for sceptics (24%), where low trust is a significant barrier to adoption.

Design for hybrid control models. The pragmatist cluster (14%) and low mean automation preference across the full sample indicate that a significant portion of consumers will only engage with AI agents if they retain meaningful oversight. Businesses should offer human-in-the-loop configurations, AI-generated shortlists with final purchase confirmation by the user, rather than fully autonomous execution. The sceptic segment requires targeted interventions including social proof mechanisms, third-party trust certifications, and gradual exposure through low-stakes categories such as digital services and low-cost electronics.

5.4 Limitations and Future Research

Several limitations of this study must be acknowledged.

The first is that the sample used was a convenience sample that was heavily weighted towards younger, more educated, and more familiar with AI, and therefore the generalisability of the results to the full population of consumers with lower digital literacy and older age profiles is limited. Future studies should replicate this study using a more representative sample of consumers that includes participants over 50, and those with less digital literacy, to determine if the main effects found in the present study, Perceived Usefulness and Trust, persist across different demographic groups.

Second, there was questionable internal reliability of the Automation Preference measure ($\alpha = 0.671$). Future research should create a more robust multi-item measure of automation preference, possibly differentiating between comfort with automation at the recommendation stage and comfort with automation at the transaction execution stage. According to previous research, these two stages represent distinct psychological processes (Schumacher et al., 2026; Castelo et al., 2019).

Third, due to the cross-sectional nature of this study, it is not possible to draw causal inferences from the data. Future research that uses a longitudinal design to track participants' intentions to adopt AI shopping agents before and after they have experienced AI shopping agents first-hand would yield stronger evidence of causality and allow for evaluation of how trust develops through use.

Fourth, the large difference between the Random Forest training R^2 value (0.853) and test R^2 value (0.174) represents a moderate level of overfitting in the Random Forest model when using a sample size of $N = 200$. A larger sample size would allow for more stability in non-linear estimates, potentially identifying interaction effects, such as H5, that cannot be observed in the current data.

Finally, the Spanish consumer context, while relevant and underexplored, may not fully reflect adoption dynamics in other cultural settings where AI regulation, consumer protection frameworks, and digital infrastructure differ substantially. Cross-cultural comparative studies, particularly contrasting European and Asian markets, represent a promising avenue for future research.

5.5 Conclusions

This study examines how behavioural, psychological, and technological factors influence consumers' willingness to adopt AI-based shopping agents, an increasingly important question given forecasts of between \$3 and \$5 trillion in global transactions through agentic commerce by 2030 (Schumacher et al., 2026). Using an extended TAM framework with a sample of 200 consumers from Spain and a multi-method analytical approach, this study yields valuable insights for retailers seeking to develop agentic commerce environments.

Perceived usefulness was found to be the strongest predictor of adoption, followed by trust. Both constructs account for almost all of the predictive power across both linear and machine-learning models. Perceived risk, previously considered a primary barrier to adoption, was not statistically significant in the multivariate model. While risk concerns do exist, they are secondary to the real barriers retailers face: insufficient demonstration of value and inadequate investment in building trust. Consumers are willing to delegate purchasing authority to an AI-based shopping system when they perceive that it acts in their best interest.

The findings indicate that Spain's consumer landscape is not monolithic. Three distinct segments emerge within the Spanish market for AI services: Enthusiasts (62%) represent a key market segment for early adoption; Pragmatists (14%) present opportunities for hybrid-control design patterns that maintain user agency; and Sceptics (24%) require a primary focus on trust-building through explainability, transparency, and gradual exposure before functional adoption can take place.

The overall insights from this study are clear: businesses must invest in demonstrating value, implementing transparent systems, and designing for the entire spectrum of consumer readiness, from Enthusiast early adopters to Sceptics, by providing the time and tools necessary to develop trust in AI-driven commerce environments.

6. References

- Abarna, K. R., Vijayalakshmi, V., & Andal, V. (2023). Impact of risk and trust on the purchase decision of online shopping customers: A study with reference to Chennai City. *International Journal of Professional Business Review*, 8(4), e01150.
<https://openaccessojis.com/JBReview/article/view/1150>
- Bain & Company. (2025). El comercio con agentic AI depende de la confianza del consumidor.
<https://www.bain.com/es/insights/agentic-ai-commerce-hinges-on-consumer-trust/>
- Bhat, S. A., & Darzi, M. A. (2020). Online service quality determinants and e-trust in internet shopping: A psychometric approach. *Vikalpa*, 45(4), 207–222.
<https://journals.sagepub.com/doi/10.1177/02560909211012806>
- Castelo, N., Bos, M. W., & Lehmann, D. R. (2019). Task-dependent algorithm aversion. *Journal of Marketing Research*, 56(5), 809–825.
<https://journals.sagepub.com/doi/10.1177/0022243719851788>
- Cuesta-Valiño, P., Núñez-Barriopedro, E., Loranca-Valle, C., & Gutiérrez-Rodríguez, P. (2025). El impacto de los agentes de IA en las decisiones de compra, la satisfacción y la felicidad del consumidor. ResearchGate.
https://www.researchgate.net/publication/394105214_EL_IMPACTO_DE_LOS_AGENTES_DE_IA_EN_LAS_DECISIONES_DE_COMPRA_LA_SATISFACCION_Y_LA_FELICIDAD_DEL_CONSUMIDOR
- Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319–340. <https://www.jstor.org/stable/249008>
- Deusto Formación. (2025). Agente de compras: qué es y cómo funciona.
<https://www.deustoformacion.com/blog/blog-logistica/agente-compras>
- Dogbe, C. S. K., Zakari, M., & Pesse-Kumar, A. G. (2019). Perceived online risk, consumer trust and M-shopping behaviour. *e-Journal of Social & Behavioural Research in Business*, 10(1), 10–23.
https://www.researchgate.net/publication/334249266_Perceived_Online_Risk_Consumer_Trust_and_M-Shopping_Behaviour

Euronews. (2025). Muchas personas usan la IA con regularidad pese a desconfiar de ella. <https://es.euronews.com/next/2025/05/09/muchas-personas-usan-la-ia-con-regularidad-pese-a-desconfiar-de-ella-segun-una-encuesta>

Featherman, M. S., & Pavlou, P. A. (2003). Predicting e-services adoption: A perceived risk facets perspective. *International Journal of Human-Computer Studies*, 59(4), 451–474. <https://www.sciencedirect.com/science/article/abs/pii/S1071581903001113?via%3Dihub>

Frank, B., & Otterbring, T. (2026). Consumer acceptance of high-autonomy AI assistants is driven by perceived benefits in online shopping settings characterized by scarcity. *Psychology & Marketing*. <https://onlinelibrary.wiley.com/doi/10.1002/mar.70074>

Ibadi, M. S., & Supriyono. (2025). The influence of price perception and trust on online shopping purchase intention in Tokopedia e-commerce. *International Journal of Economics*, 4(2), 1446–1454. <https://ejournal.ipinternasional.com/index.php/ijec/article/view/1685>

Iglesias, O. (2025). Radiografía de las opiniones sobre la inteligencia artificial en España. Fundación Sistema. <https://fundacionsistema.com/radiografia-de-las-opiniones-sobre-la-inteligencia-artificial-en-espana/>

Innovación Digital 360. (2025, January 19). Los agentes de IA ya están comprando por tus clientes (y tu ecommerce no lo sabe). <https://www.innovaciondigital360.com/cio/agentes-ia-en-ecommerce/>

InsiderOne. (2025). Cómo los agentes de compras con IA impulsan las conversiones y los ingresos. <https://insiderone.com/es/ai-shopping-agents-drive-conversions-revenue/>

Kim, D. J., Ferrin, D. L., & Rao, H. R. (2008). A trust-based consumer decision-making model in electronic commerce. *Decision Support Systems*, 44(2), 544–564. <https://www.sciencedirect.com/science/article/abs/pii/S0167923607001005?via%3Dihub>

Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, 46(1), 50–80. <https://scispace.com/pdf/trust-in-automation-designing-for-appropriate-reliance-2uiy4o89ga.pdf>

León, C. (2025). Los pagos impulsados por agentes de IA buscan redefinir el e-commerce y la prevención del fraude. *Forbes México*.

<https://forbes.com.mx/los-pagos-impulsados-por-agentes-de-ia-buscan-redefinir-el-e-commerce-y-la-prevencion-del-fraude/>

Luo, X., & Zhang, Y. (2025). How do consumers trust and accept AI agents? An extended theoretical framework and empirical evidence. *Behavioral Sciences*, 15(3), 337.

<https://www.mdpi.com/2076-328X/15/3/337>

Mani, S., Tiwari, P., Ramchandani, S., Acharya, P. S., & Irudayaraj, V. D. (2025). From clicks to conversions: How AI shapes consumer trust, experience, and online buying behaviour. *Advances in Consumer Research*, 2(4), 5028–5035.

<https://acr-journal.com/article/from-clicks-to-conversions-how-ai-shapes-consumer-trust-experience-and-online-buying-behaviour-1612/>

Qalati, S. A., Galvan Vela, E., Li, W., Dakhan, S. A., Thuy, T. T. H., & Merani, S. H. (2021). Effects of perceived service quality, website quality, and reputation on purchase intention: The mediating and moderating roles of trust and perceived risk in online shopping. *Cogent Business & Management*, 8, 1869363.

<https://www.tandfonline.com/doi/full/10.1080/23311975.2020.1869363>

Rasty, F., Mirghafoori, S. H., Saeida Ardekani, S., & Ajdari, P. (2021). Trust barriers to online shopping: Investigating and prioritizing trust barriers in an intuitionistic fuzzy environment. *International Journal of Consumer Studies*, 45(5), 1030–1046.

<https://onlinelibrary.wiley.com/doi/full/10.1111/ijcs.12629>

Riskified. (2025). Global study: 73% of shoppers using AI in shopping journey — but merchants face new agentic commerce risks.

<https://www.riskified.com/press/global-study-73-of-shoppers-using-ai-in-shopping-journey-but-merchants-face-new-agentic-commerce-risks/>

Schumacher, K., Keutel, M., & Kluge, P. (2026). Europe's agentic commerce moment: Decision influence is here; execution is coming. McKinsey & Company.

<https://www.mckinsey.com/capabilities/quantumblack/our-insights/europes-agentic-commerce-moment-decision-influence-is-here-execution-is-coming>

Shin, D. (2021). The effects of explainability and causability on perception, trust, and acceptance: Implications for explainable AI. *International Journal of Human-Computer Studies*, 146, 102551.

<https://www.sciencedirect.com/science/article/abs/pii/S1071581920301531?via%3Dihub>

Silverfort. (2025). Beyond the hype: The hidden security risks of AI agents and MCP.
<https://www.silverfort.com/es/blog/beyond-the-hype-the-hidden-security-risks-of-ai-agents-and-mcp/>

ScienceDirect. (2024). Artificial intelligence and organizational agility: An analysis of scientific production and future trends
<https://www.sciencedirect.com/science/article/pii/S2444883424000135>

Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*, 27(3), 425–478.
https://www.researchgate.net/publication/220259897_User_Acceptance_of_Information_Technology_Toward_a_Unified_View

7. Appendix

7.1 Survey

<https://forms.gle/uE6tod9N6QPa6EWA9>

7.2 Methodology

Table A1. Variables and Measurement

Variable	Type	Survey Items	Role in Model
Perceived Usefulness	Independent	PU1, PU2, PU3	H1 (expected +)
Trust	Independent	TR1, TR2, TR3, TR4	H2 (expected +)
Perceived Risk	Independent	PR1, PR2, PR3, PR4	H3 (expected –)
Automation Preference	Independent	AP1, AP2, AP3 (reverse coded)	H4 (expected +)
Trust × Perceived Risk	Interaction	Mean-centered product term	H5 (moderation)
Adoption Intention	Dependent	AI1, AI2, AI3	Outcome variable (Y)
Age, Gender, Education, AI Familiarity, Digital Experience, Online Shopping Frequency	Control	Demographic & behavioural items	Moderators

7.3 Results

Table A2. Descriptive Statistics: Composite Variables

Variable	M	SD	Min	Max	N
Perceived Usefulness	3.557	0.966	1.00	5.00	200
Trust	3.167	0.986	1.00	5.00	200
Perceived Risk	3.761	0.858	1.00	5.00	200
Automation Preference	1.677	0.810	1.00	5.00	200
Adoption Intention	3.472	1.018	1.00	5.00	200

Table A3. Pearson Correlation Matrix

Variable	1. PU	2. TR	3. PR	4. AP	5. AI
1. Perceived Usefulness	—				
2. Trust	.770***	—			
3. Perceived Risk	-.103	-.153*	—		
4. Automation Preference	-.196**	-.007	-.444***	—	
5. Adoption Intention	.714***	.686***	-.146*	-.160*	—

Note. * $p < .05$ ** $p < .01$ *** $p < .001$. PU = Perceived Usefulness; TR = Trust; PR = Perceived Risk; AP = Automation Preference; AI = Adoption Intention.

Table A4. Random Forest: Feature Importance vs. MLR Standardised Betas

Variable	RF Importance	MLR std. β	RF Rank	MLR Rank
Perceived Usefulness	52.5%	0.392	1	1
Trust	31.3%	0.364	2	2
Perceived Risk	10.1%	0.099	3	4
Automation Preference	6.1%	0.126	4	3

Note. RF Train $R^2 = .853$, Test $R^2 = .174$, CV R^2 (5-fold) = .489 (SD = .067). MAE = 0.619, RMSE = 0.878.

Table A5. K-Means Cluster Profiles: Mean Scores per Variable

Variable	Enthusiasts (n=124, 62%)	Pragmatists (n=29, 14%)	Sceptics (n=47, 24%)	Sample M	Sample SD
Perceived Usefulness	4.056	3.034	2.560	3.557	0.966
Trust	3.607	3.052	2.080	3.167	0.986
Perceived Risk	3.821	2.655	4.287	3.761	0.858
Automation Preference	1.457	3.103	1.376	1.677	0.810
Adoption Intention	4.019	3.092	2.262	3.472	1.018

Note. Mean scores on 1–5 Likert scale. Silhouette score ($K=3$) = 0.308.

